DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. **Branch: Computer Engineering** **Semester: VIII**

Subject Code & Name: BTCOE802 (B) Cryptography & Network Security

Max Marks: 60 **Date: 07/07/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	What is Cryptography? Explain The main aspects of cryptography.	Remember	06
B)	Explain What is Block Cipher? How does Block Cipher Works?	Remember	06
C)	Illustrate Remainders and Modular Arithmetic.	Apply	06
Q.2	**Solve Any Two of the following.**		
A)	Find the Modular Inverse **i.** A=3, M=7 **ii.** A=2 M=6 **iii.** A=3 M=11	Evaluate	06
B)	Prove Fermat's little theorem states. if p is a prime number, then for any integer a, the number $a^p - a$ is an integer multiple of p. $a^p \equiv a \pmod p$. Special Case: If a is not divisible by p, Fermat's little theorem is equivalent to the statement that $a^{p-1}-1$ is an integer multiple of p. Here a is not divisible by p ap-1 ≡ 1 (mod p) OR ap-1 % p = 1	Apply	06
C)	Differentiate Symmetric vs. Asymmetric Cryptography	Analyze	06
Q. 3	**Solve Any One of the following.**		
A)	What is ElGamal Cryptosystem? Write code for implementing ElGamal Cryptosystem	Evaluate	12
B)	What is Rabin Cryptosystem? Explain steps in Rabin Cryptosystem.	Analyze	12
C)	Explain in details the classification Process types of Digital Signature.	Apply	12
Q.4	**Solve Any Two of the following.**		
A)	Explain Working of Cryptographic Hash Function.	Analyze	06
B)	Describe SHA Forms.	Apply	06
C)	How does the Digital Signature Standard work?	Apply	06
Q. 5	**Solve Any One of the following.**		

A)	Explain Cryptanalysis techniques for stream ciphers.	Remember	12
B)	What is secret sharing? Explain with Example. Also Explain how Shamir's Secret Key Works?	Evaluate	12
C)	Describe Types of Side Channel Attack.	Analyze	12

***** End *****